

Priority

DL Architectures

Classical CNN Architectures

- LeNet
- AlexNet
- VGG
- GoogLeNet / Inception
- ResNet
- DenseNet
- MobileNet
- EfficientNet

Recurrent Architectures

- RNN
- LSTM
- GRU

Transformers & Variants

- Transformer
- BERT
- GPT (GPT-1, GPT-2, GPT-3, GPT-4)
- T5
- LLaMA
- Mistral
- Gemma

Vision Transformers (ViT)

- ViT
- Swin Transformer
- DeiT

Autoencoders and Variants

- Autoencoder (AE)

- Variational Autoencoder (VAE)
- VQ-VAE

Generative Adversarial Networks (GANs)

- GAN
- DCGAN
- CycleGAN
- StyleGAN
- BigGAN

Diffusion Models

- DDPM (Denoising Diffusion Probabilistic Models)
- Latent Diffusion Models (LDM)
- Denoising Score Matching

Library-Specific

PyTorch Core

- Tensors & Autograd
- `nn.Module`, `forward()`
- Optimizers (SGD, Adam, etc.)
- Loss Functions (`CrossEntropyLoss`, `MSELoss`, etc.)
- CUDA / Device Management
- Hooks & Callbacks

Custom Dataloaders

- `Dataset` & `DataLoader` classes
- `__getitem__()`, `__len__()`
- `collate_fn`
- Iterable Datasets
- Multi-worker Loading
- Custom Transforms / Augmentations

Multi-node Training

- `torchrun` CLI tool
- Environment variables for multi-node setup

- `torch.distributed.launch` (legacy)
- Master address, rank, world size config

Distributed Training (5D Parallelism)

- **Data Parallelism** → `DistributedDataParallel` (DDP)
- **Tensor Parallelism (TP)** → Partitioning model tensors (e.g., Megatron-LM)
- **Pipeline Parallelism (PP)** → Splitting model layers across devices
- **Expert Parallelism** → Mixture-of-Experts (e.g., Switch Transformers)
- **Sequence Parallelism** → For extremely long sequences (e.g., FlashAttention)
- Gradient Accumulation
- Activation Checkpointing

Deepspeed

- Zero Redundancy Optimizer (ZeRO-1/2/3)
- ZeRO-Offload
- Activation Partitioning
- Mixed Precision (FP16/BF16)
- Sparse Attention
- `deepspeed.initialize` and config JSONs
- Training large models with 3D parallelism

Accelerate (Hugging Face)

- Lightweight distributed training wrapper
- Supports multi-GPU, TPU, multi-node setups
- Integration with Transformers & Datasets
- Handles device placement, mixed precision
- `accelerate launch`, `accelerate config`
- Checkpointing and logging built-in

Transformers (Hugging Face)

- `AutoModel`, `AutoTokenizer` abstractions
- Pretrained model hub
- Trainer / TrainingArguments
- Token classification, QA, generation pipelines
- Integration with Accelerate, Deepspeed, PEFT, etc.
- ONNX / TorchScript export

vLLM

- Fast, memory-efficient LLM inference engine

- Continuous batching / paged attention
- Supports OpenAI API compatibility
- Token streaming & multi-turn chat
- Efficient KV cache management
- Designed for high-throughput serving

Sglang Internal

- Declarative LLM workflow DSL
- Token-by-token streaming
- Prompt caching and reuse
- Built-in function/tool calling
- Dynamic control flow in LLM apps
- Easy integration with vLLM backend

VLM Specific

Foundational Contrastive VLMs

- **CLIP** (Contrastive Language–Image Pretraining) – OpenAI
- **SigLIP** – Sigmoid loss variant of CLIP (Google)
- **DINO / DINOv2** – Self-supervised ViT pretraining with some VLM crossover
- **MetaCLIP** – Contrastive model using curated web data
- **UniCL** – Unified contrastive learning
- **Coca** – Contrastive Captioners (Google)

Captioning + Generation-Centric Models

- **BLIP** – Bootstrapped Language–Image Pretraining
- **BLIP-2** – Two-stage model with Q-former and frozen LLM
- **Pix2Struct** – Vision-to-text for structured documents (Google)
- **SimVLM** – Simple Vision-Language Model (Google)
- **Flamingo** – Perceiver-resampler + decoder LM (DeepMind)
- **MiniGPT-4** – Vision encoder + Vicuna for chat
- **InstructBLIP** – Instruction-tuned VLM
- **PaLI / PaLI-X** – Multilingual, multimodal vision-language models
- **GIT** – Unified image-text generation (Microsoft)
- **OFA** – Unified framework for vision + language tasks

Chat-Centric Multimodal LLMs

- **LLaVA** – LLaMA + CLIP-ViT + Chat interface

- **LLaVA-1.5 / LLaVA-Next** – Improvements in training & alignment
- **Qwen-VL / Qwen-VL-Chat** – Alibaba’s multimodal extension of Qwen
- **mPLUG-Owl / Owl-ViT** – Open-ended VQA and captioning
- **CogVLM** – Cognition-injected VLM (THUDM)
- **MiniCPM-V / MiniCPM-Llama3-V** – Vision + tiny LLM (OpenCalm)
- **InternLM-XComposer / InternLM2 / InternVL-Chat** – Visual reasoning + LLM
- **Phi-3 Vision** – Small, multimodal Phi-3 from Microsoft
- **InternVL** – Vision-language branch of InternLM
- **Florence-2** – Unified V+L understanding and generation (Microsoft)
- **Gemma-3** – Multimodal variant of Google’s Gemma

Miscellaneous

- **SAM (Segment Anything Model)** – Foundation model for universal segmentation, often used for grounding and region-based VLMs
- **DETR / GLIP / GIT / GRIT** – Object detectors and grounded image-text models
- **OWL-ViT / GroundingDINO** – Open-vocabulary detection and grounding

VLM Techniques & Components

Vision Encoding Techniques

- **Q-Former** – Query-based transformer to extract relevant visual features (BLIP-2)
- **Perceiver Resampler** – Compresses visual tokens via latent cross-attention (Flamingo)
- **Visual Token Pruning** – Selects top-K informative tokens to reduce computation
- **Token Merging / Token Pooling** – Merges similar visual tokens to reduce sequence length
- **Fused Patches** – Merging high-res visual features for global-local representation

Resolution & Memory Optimization

- **H-Reducer (Horizontal Reducer)** – Linearly projects wide spatial features to reduce width dimension; helps with high-res input compression
- **Pixel Shuffle / Unshuffle** – Rearranges spatial resolution and channel depth (used in super-resolution and some high-res input downsampling)
- **Patch Merging** – Used in ViT/transformers to reduce spatial dimensions gradually
- **Spatial Feature Pooling** – Mean/max pooling across H/W dimensions
- **Multi-Scale Features** – Hierarchical features from CNN backbones or Swin-style ViTs

Input Conditioning & Representation

- **Bounding Box Embedding** – Adds spatial info for object-aware reasoning (e.g., GRIT, GIT)

- **Region Features** – Extracted from Faster R-CNN, DETR, etc., and injected into transformer
- **Learned Queries** – Used to attend to specific visual concepts (Q-former, DETR-like models)
- **Scene Graphs / Layout Encoding** – Structured input to represent spatial relationships
- **Visual Prompt Tuning** – Learnable tokens that adapt frozen vision encoders

Multimodal Pretraining Objectives

- **ITC (Image-Text Contrastive)** – Aligns image and text embeddings
- **ITM (Image-Text Matching)** – Binary task of paired vs unpaired samples
- **MLM (Masked Language Modeling)** – Language modeling with masked tokens
- **MIM (Masked Image Modeling)** – Masked patch prediction (BEiT, MAE)
- **VQA Supervision** – Question answering over visual inputs
- **Caption Generation** – Supervised image/text generation
- **Referential Grounding** – Predict box/span given referring expressions

Training & Alignment Techniques

- **Visual Instruction Tuning** – Aligning VLMs to follow multimodal instructions
- **LoRA / QLoRA for Vision Modules** – Adapter-based efficient tuning
- **Vision-Language Alignment Losses** – E.g., contrastive loss, similarity loss
- **Cross-modal Attention** – Language attends to visual features (or vice versa)
- **Co-attention Mechanisms** – Mutual attention across modalities

Other Topics

ML Fundamentals

Core Concepts

- Supervised Learning
- Unsupervised Learning
- Semi-supervised Learning
- Reinforcement Learning
- Online Learning
- Transfer Learning
- Self-supervised Learning

Models & Algorithms

- Linear Regression
- Logistic Regression

- Decision Trees
- Random Forests
- **Gradient Boosting (XGBoost, LightGBM, CatBoost)**
- k-Nearest Neighbors (k-NN)
- Support Vector Machines (SVM)
- Naive Bayes
- Clustering (k-Means, DBSCAN, Hierarchical)
- Hidden Markov Models

Model Evaluation

- Train/Test Split
- Cross-Validation
- Bias-Variance Tradeoff
- Overfitting / Underfitting
- Precision, Recall, F1-Score
- **ROC Curve, AUC**
- Confusion Matrix
- Log Loss, Hinge Loss, MAE, MSE, RMSE

Optimization

- Gradient Descent
- Stochastic Gradient Descent (SGD)
- **Momentum, Nesterov**
- Adam, RMSProp, Adagrad
- **Learning Rate Scheduling**

Feature Engineering

- One-hot Encoding
- Label Encoding
- Feature Scaling (Normalization, Standardization)
- Binning
- Polynomial Features
- **Dimensionality Reduction (PCA, t-SNE, UMAP)**

Regularization

- L1 Regularization (Lasso)
- L2 Regularization (Ridge)
- Elastic Net

Uncertainty & Probabilistic Modeling

- Probability Distributions
- Bayes' Theorem
- Maximum Likelihood Estimation
- MAP Estimation

DL Fundamentals

Core Concepts

- Perceptron
- Activation Functions (ReLU, Sigmoid, Tanh, GELU)
- Feedforward Neural Networks (MLP)
- Backpropagation
- Loss Functions (Cross-Entropy, MSE, MAE)
- Epoch, Batch Size, Iteration
- Vanishing/Exploding Gradients
- **Weight Initialization (Xavier, He)**
- Dropout
- Batch Normalization
- **Layer Normalization**
- Skip Connections / Residual Learning
- Attention Mechanism
- Positional Encoding

Training Techniques

- Gradient Descent & Variants (SGD, Adam, RMSProp)
- **Learning Rate Scheduling (Step, Cosine, Warmup)**
- **Early Stopping**
- Data Augmentation
- Regularization (L1, L2, Dropout)
- Transfer Learning
- Fine-tuning vs. Feature Extraction
- Pretraining

DL for Sequences

- Sequence-to-Sequence Models
- Encoder-Decoder Architectures
- Attention-based Models
- Causal Masking (for autoregression)

DL for Vision

- Convolutions
- Receptive Field
- Image Classification
- **Object Detection (Faster R-CNN, YOLO, SSD)**
- Segmentation (U-Net, DeepLab)

DL for NLP

- Word Embeddings (Word2Vec, GloVe)
- Positional Embeddings
- Transformer-based LMs (BERT, GPT)
- Pretraining Objectives (MLM, CLM, NSP)
- **Tokenization (BPE, WordPiece, SentencePiece)**

DL Tooling

- Frameworks (TensorFlow, PyTorch, JAX)
- GPU/TPU Acceleration
- Mixed Precision Training
- Model Checkpointing
- Logging & Visualization (TensorBoard, Weights & Biases)

Computer Vision

Fundamentals

- Image Representation (RGB, Grayscale, Channels)
- Convolution & Filters (Edge Detection, Sharpening, Blurring)
- Histogram Equalization
- Thresholding (Global, Adaptive, Otsu's)
- Morphological Operations (Erosion, Dilation, Opening, Closing)
- Color Spaces (RGB, HSV, Lab, YCbCr)
- Resizing, Cropping, Padding
- Image Pyramids
- **Fourier Transform (DFT, FFT)**

Classical Computer Vision

- Contour Detection
- Edge Detection (Sobel, Canny)
- Corner Detection (Harris, Shi-Tomasi)
- Feature Descriptors (SIFT, SURF, ORB)
- Feature Matching (BFMatcher, FLANN)
- Optical Flow (Lucas-Kanade, Farneback)

- Background Subtraction
- Homography & Image Stitching
- Camera Calibration
- Epipolar Geometry
- Structure from Motion
- SLAM (Simultaneous Localization and Mapping)

Image Segmentation

- Thresholding-Based
- Region-Based (Watershed, Region Growing)
- Clustering-Based (k-Means)
- Graph-Based
- Deep Learning-Based (U-Net, Mask R-CNN, DeepLab)

Object Detection

- Classical (Haar Cascades, HOG + SVM)
- CNN-Based:
 - R-CNN
 - Fast R-CNN
 - Faster R-CNN
 - YOLO (You Only Look Once)
 - SSD (Single Shot Detector)
 - RetinaNet
 - DETR (DEtection TRansformer)

Image Classification

- CNN Architectures (AlexNet, ResNet, EfficientNet, ViT)
- Transfer Learning
- Fine-Tuning

Image Generation & Restoration

- Denoising
- **Super-Resolution (SRCNN, ESRGAN)**
- Inpainting
- **Style Transfer**
- Image-to-Image Translation (Pix2Pix, CycleGAN)
- Generative Models (GANs, Diffusion Models)

Face & Gesture Recognition

- **Face Detection (Viola-Jones, MTCNN, RetinaFace)**

- Face Recognition (FaceNet, ArcFace, DeepFace)
- Facial Landmark Detection
- Gesture Recognition

3D Vision

- Depth Estimation
- Stereo Vision
- Point Clouds
- 3D Reconstruction
- LiDAR Processing

Tools & Libraries

- OpenCV
- PIL / Pillow
- skimage (scikit-image)
- imgaug / Albumentations (Augmentation)
- MediaPipe
- PyTorch/TensorFlow (for deep learning pipelines)

NLP

Core Concepts

- **Tokenization (Word, Subword, Sentence)**
- Lemmatization & Stemming
- Stopword Removal
- Part-of-Speech (POS) Tagging
- Named Entity Recognition (NER)
- Dependency Parsing
- Constituency Parsing
- **Chunking**
- **N-grams**
- Language Modeling
- Word Sense Disambiguation
- Text Normalization (Lowercasing, Punctuation Removal)

Classical NLP

- Bag of Words (BoW)
- TF-IDF
- Rule-based Systems

- Regular Expressions
- Statistical Language Models (n-gram models)
- **Latent Semantic Analysis (LSA)**
- Latent Dirichlet Allocation (LDA)

Embeddings & Representations

- One-Hot Encoding
- Word Embeddings (Word2Vec, GloVe, FastText)
- Contextual Embeddings (ELMo, BERT, GPT)
- Sentence Embeddings (Sentence-BERT, USE)
- Positional Embeddings

NLP Tasks

- Text Classification (Spam Detection, Sentiment Analysis)
- Sequence Labeling (POS, NER)
- Named Entity Recognition
- Machine Translation
- Text Summarization (Extractive, Abstractive)
- Question Answering
- Text Generation
- Conversational AI / Chatbots
- Natural Language Inference (NLI)
- **Coreference Resolution**
- Information Extraction
- Relation Extraction
- Semantic Role Labeling

Transformer-Based Models

- BERT
- RoBERTa
- GPT (1 to 4)
- XLNet
- T5
- ELECTRA
- DistilBERT
- ALBERT
- LLaMA
- Mistral
- Falcon

Pretraining Objectives

- MLM (Masked Language Modeling)
- CLM (Causal Language Modeling)
- NSP (Next Sentence Prediction)
- SOP (Sentence Order Prediction)
- Denoising Objectives (T5-style)

Evaluation Metrics

- Accuracy, Precision, Recall, F1
- BLEU (Translation)
- ROUGE (Summarization)
- METEOR
- **Perplexity**
- Exact Match (EM)
- **NDCG / MRR (Ranking)**

Libraries & Tools

- NLTK
- spaCy
- Gensim
- Hugging Face Transformers
- OpenNLP
- Stanford CoreNLP
- AllenNLP
- Flair

RAG

Core IR Concepts

- Document Collection / Corpus
- Indexing (Inverted Index, Dense Index)
- Tokenization & Normalization
- Stemming & Lemmatization
- Stopword Removal
- TF-IDF
- **BM25 (Okapi BM25)**
- Vector Space Model
- Probabilistic Models (e.g., Query Likelihood Model)

- Query Expansion
- Relevance Feedback

Neural IR

- Dense Representations (Sentence Transformers, DPR)
- Dual Encoders (Bi-encoders)
- Cross Encoders
- Late Interaction (ColBERT)
- SPLADE / Sparse Neural Representations
- Hard Negative Mining
- Approximate Nearest Neighbor Search (ANN)
 - **FAISS**
 - ScaNN
 - **HNSW**

IR Evaluation Metrics

- Precision@K
- Recall@K
- F1-Score
- Mean Reciprocal Rank (MRR)
- Mean Average Precision (MAP)
- Normalized Discounted Cumulative Gain (NDCG)
- Hit Rate

Retrieval-Augmented Generation (RAG)

Core RAG Concepts

- Retrieve-then-Generate Paradigm
- Fusion-in-Decoder vs Fusion-in-Encoder
- Top-K or Top-N Retrieval
- Contextual Document Selection
- Chunking & Sliding Window Techniques
- Query Rewriting (e.g., in conversational RAG)

RAG Architectures

- RAG (Facebook AI)
- FiD (Fusion-in-Decoder)
- REALM (Google)
- RETRO (DeepMind)
- Atlas (Meta)

- HyDE (Hypothetical Document Embeddings)
- LLM + Vector DB Pipelines (e.g., LangChain, LlamaIndex)

Components of a RAG System

- **Retriever**
 - BM25
 - Dense Retrievers (DPR, Sentence-BERT)
 - Hybrid Retrieval (Sparse + Dense)
- **Generator**
 - T5
 - BART
 - GPT-style LMs
 - LLaMA/Mistral/Claude in retrieval-augmented setups
- **Memory Store**
 - FAISS
 - Elasticsearch
 - Milvus
 - Pinecone
 - Weaviate

Applications

- Open-domain Question Answering
- Conversational Agents
- Knowledge-augmented Chatbots
- Legal/Biomedical QA
- Enterprise Search
- LLM Fact Grounding

Tools & Frameworks

- Haystack
- LangChain
- LlamaIndex
- Hugging Face Transformers
- FAISS / ScaNN
- Pyserini / Anserini
- Elastic / Vespa

LLM-centric Topics

Parameter-Efficient Fine-Tuning (PEFT)

- **LoRA (Low-Rank Adaptation)**
- **QLoRA (Quantized LoRA)**
- Prefix Tuning
- Prompt Tuning
- P-Tuning / P-Tuning v2
- Adapter Layers
- BitFit (Bias Fine-Tuning)
- IA³ (Infused Adapter by Inhibiting and Amplifying Attention)

Quantization

- **Post-Training Quantization (PTQ)**
- Quantization-Aware Training (QAT)
- INT8 / INT4 Quantization
- **GPTQ (Quantized Weight-Only PTQ)**
- **AWQ (Activation-Aware Weight Quantization)**
- BitsAndBytes (4-bit / 8-bit inference)
- SmoothQuant (Activation-aware PTQ for LLMs)

Model Pruning & Compression

- Weight Pruning (magnitude, structured, unstructured)
- Neuron Pruning
- Head Pruning (for attention)
- Knowledge Distillation
- Teacher-Student Training

Efficient Inference

- KV Cache / Prefill + Decode Separation
- PagedAttention (vLLM)
- **FlashAttention / FlashAttention-2**
- **Grouped Query Attention (GQA)**
- **Multi-Query Attention (MQA)**
- **Speculative Decoding**
- Continuous Batching / Streaming Generation

Multi-modal & Sparse Architectures

- Mixture of Experts (MoE)
- Sparse Mixture Routing
- Gated Linear Units (GLUs)
- Switch Transformer
- Router Tokens

Memory Optimization

- Activation Checkpointing
- **Gradient Accumulation**
- **ZeRO (Zero Redundancy Optimizer)**
- Offloading (CPU/Flash)
- FP16 / BF16 / Mixed Precision